



Greenhouse Cucumbers

CULTIVARS

- Kalunga (English)
- Katrina (Lebanese)

INTENSIVE SPACING

- English: 1 row per bed spaced every 18" (50 cm) on the row.
- Lebanese: 1 row per bed spaced every 12" (33 cm) on the row.
- These are parthenocarpic varieties (meaning they don't need to be pollinated) planted on 30" beds (75 cm) with 12" (5 cm) pathways and trellised in a modified umbrella style. Final density for English cukes is 1.8 plants/m². For Lebanese cukes, final density is 3 plants/m². See **Density Appendix** for explanations.

FERTILIZATION

- Bimonthly application of compost (2-1-1), alfalfa meal (2-0-2), feather meal (11-0-0) Sul-Po-Mag (0-0-22) and potassium sulfate (0-0-50). See **Fertilization Appendix** for details.
- Perform an EC (electrical conductivity) test every 2 weeks to monitor the progression of the fertilization. The EC will determine if we need to fertilize or not.
- We fertilize once on one side of the row, then on the other side 2 weeks later. The EC test is to be conducted before fertilization on the side that is meant to be fertilized that week.

TRANSPLANTING AND SET UP

The bed layout when planting cucumbers is the same as for all other crops grown in our greenhouses. Beds are 42" (1 m) from centre to centre—beds are 30" (75 cm) and pathways 12" (5 cm). The cucumbers are trellised onto 1 wire which runs parallel to

the center of the 30" bed. The height of the wire is +/-7' (2M) and it should be at least 12 gauge. See **Crop Spacing and Hardware Set-Up Appendix** for details.

Before transplanting:

Make sure that the steel support wire is installed and ready to go before you set out your transplants.

The success of a cucumber transplant is largely a result of both soil and air temperatures. **Soil temperatures have to be above 62 °F (17 °C)** and the ambient temperature (day and night) should be above 61 °F (16 °C). Failure to maintain the plants to such minimums will shut the plants down and adversely affect the root uptake. Cucumber transplants are sensitive, and should never be allowed to become stressed for water or nutrients. A bad start is a set-up for future problems.

Make sure all furnaces are operational, with back-up ready to go. Plan to measure soil temperatures 2 weeks prior to the transplanting date. If too cold, cover beds with clear plastic—a few sunny days will help warm up the soil. In cold climates, installing radiant floor heating (PEX tubing buried 18" in the ground + simple household hot water tank) is another solution.

During those 2 weeks, it is also good practice to flood the soil of the greenhouse with 1-2 inches (2-3 cm) of water. This slowly penetrating water will flush out salts (left from fertilizers over previous years) and increase the conductivity of the warming soil.

IMPLANTATION

Broadfork the beds.

Mark the spacings every 12" or 18", then dig the holes using a round shovel (+/-5" diameter). Add some compost in each hole and a handful of chicken manure (5-2-1) or other fast-acting balanced natural fertilizer. Then, fill the hole with water using a hose. The mixture should become a muddy matter.

On our first planting in the spring, when the soil is not warm, we sometimes use a rototiller and incorporate all our amendments directly in top 8 inches of the soil. Tilling will add lots of oxygen in the bed, warming it up.

Place cucumber seedling in front of each hole, making sure not to mix up the cultivars. Start by delicately extracting the seedlings from their pots and placing them gently into the hole. The soil level on the seedling should end up being level with the soil. Cucumber seedlings have sensitive roots. Handle them carefully, and don't bury their collars.

Attach one piece of twine to the plant using clips and tie it to the overhead wire. Install 4 drip tape lines, but activate only the 2 closest to the plants. This way, we avoid over-mineralizing the rest of the bed and are keeping the "pantry" abundant for another few weeks. 2 weeks later, open the 2 outside drip tape lines, bringing the total to 4 drip tape lines per 30" (75 cm) beds.

Install white on black plastic between rows. The tarp helps keep the fertilizers moist and the soil dark so that the earthworms, woodlouse and other critters may gather to feed 24 hours a day within the soil. On the flip side, its white surface serves to increase light reflection onto the plants, thereby encouraging photosynthesis.

MANAGING TEMPERATURES

For 7 days after transplanting, maintain a constant temperature (day and night) of 73 °F to 77 °F (23 °C to 25 °C). This will encourage vegetative growth and root retake. After the first week, maintain the same daytime temperature but lower the night temperatures to between 70 °F and 73 °F (21 to 24 °C)

The best way to heat your greenhouse is to install venting tubes in the pathways between each bed (see video). To help prevent fungal problems, we leave the blowers to ventilate 24/7. This is done by wiring the blower of your furnace so that it's always running.

When cucumber harvest starts, try to maintain daytime temperatures in the greenhouse between 70 °F to 75 °F (21-24°C), and night temperatures between 63 °F and 68 °F (17-20°C). A good twenty-four-hour average for cucumbers is 70 °F (21 °C). If possible, avoid temperatures above 86 °F (30 °C) and in the event of hot days, err on the side of watering too much, rather than not enough. It's OK for cucumber leaves to wilt during hot afternoons, as long as they perk back up at the end of the day. If they don't, more water or ventilation may be needed.

As for humidity, the optimal rate for greenhouse cucumbers is between 65% and 80%. An environment that is too humid will lead to problems related to fungal disease

and to poor floral fertilization. Excess or lack of humidity will strongly influence crop productivity.

To produce regularly, cukes need a great amount of light. In greenhouses where the climate is temperate, cucumbers will grow faster in summer when daylight lasts 17 to 18 hours, compared to fall when daylight decreases to 12 hours and less.

PRUNING, SUCKERING & TRELLISING

Pruning cucumber plants needs to be done **at least once a week**, but preferably twice a week. Otherwise, the plant tends to set more fruits than it can support, which ultimately leads to new fruits being aborted.

For both English and Lebanese cukes, in the first 2 weeks, or before the plant reaches 2' (60 cm) high, all the fruits should be pruned off the plant. By limiting the number of fruits on the plants while it's establishing its roots, you will have higher total yields.

After the first two weeks, follow these simple pruning guidelines:

English cukes:

Prune to leave 1 fruit every 2 nodes and remove all tendrils + suckers.

Lebanese cukes:

Leave all fruits, but remove all tendrils + suckers.

Lebanese cucumbers are more vigorous than English, hence the reason why their fruits are not removed.

In both cases, wind the plant around the twine clockwise. We prefer winding the plant around the string, instead of using clips to the trellis. It is faster and eliminates having to carry around material requiring manipulation.

Pruning and winding should be done twice a week.

When the head of the plant reaches the wire, attach it with a clip and train it to grow along the wire to its right. Until the main head reaches the next plant, let all the suckers grow—these will form new heads that will be shooting down. Once the main

leader reaches the next plant, clip it to the wire for support, and let it grow downwards.

By that time, you should have about 3 suckers that have developed into new heads. These are growing downwards (from the top wire heading towards the ground) and are treated just like the main head, pruning off fruits, tendrils and suckers every week.

When the main leader reaches down to 5 fruits (10–12 nodes), prune off its head, which will stop this stem from producing any new fruits. Production continues on the other leaders until they also reach 5 fruits, at which point, you can top them too.

This goes on as long as the plants keep producing, but ultimately after 10 to 12 weeks, production declines considerably. It is therefore time to replace the plants with a new succession. To make sure we don't have a gap in our production, we plan for this new succession to start producing 10 weeks after the plants it replaces.

PRUNING LEAVES

Similar to removing suckers, pruning leaves allows the plant's energy to go toward fruit production, instead of excess vegetative growth. Removing the lower leaves also increases airflow, which can reduce the incidence of disease.

You can begin pruning the bottom leaves when the head of the plant reaches the top of the trellising wire. At that point, the plant should have +/-18–20 leaves and you can remove 1–2 leaves every second week. The leaves most appropriate to remove are those on the lower parts of the plant that may be shaded or crowded. Additionally, if you see any leaves that appear diseased, remove these and dispose of them. As with pruning suckers, it is best done early in the morning and only when the foliage is dry, in order to avoid any potential spread of disease.

IRRIGATION

Cucumber plants require lots of water when in full production during the summer: 3 to 4 litres per plant per day. Drip tape is the best way to deliver the water right where it is needed and on a 30" (75 cm) bed, there should be 4 drip lines watering the plants. The amount will depend on the stage of growth, outside temperatures and more, but these guidelines should be followed:

Schedule 1 to 3 irrigation cycles per day (and no more)

These cycles should be done between 9 AM and 2 PM. The objective is to not water the plants 2 hours before the sun rises (giving the plants time to get active) and not have them begin the night with a wet soil. The 9 AM to 2 PM rule can be adjusted, as long as it follows those principles.

We use 3 irrigation cycles which are programmed on the timer: one each for sunny, partially sunny and cloudy days. Outside conditions necessitate different watering cycles: 3 per day when sunny, 2 per day when partially sunny and only 1 per day when cloudy. To manage this, we use a watering clock with different start times and cycles. The irrigation is adjusted each morning according to sunrise and sunset times and weather. It takes only a few seconds a day to adjust the water clock.

The length of each irrigation pulse has to be adjusted to ensure that the plants are getting the right amount of water. A good way to test this is to sample soil in the greenhouse at the end of the day and first thing the next morning—take a handful of soil at each point, and judge the moisture content. Both should feel about the same.

WEEDING

Crop is kept clean by covering the soil with UV treated plastic tarps between the rows and by occasional hand weeding in the row.

PLANT PROTECTION

Pest control is a major part of successful greenhouse cucumber growing. How to deal with the problem will depend on many different variables, but prevention and observation should be your primary tools. The aim is to favour a plant positive approach to pest management, avoiding using pesticide use when possible and working with predatory insect introduction instead. It's the way to go, but it requires preparation and advance planning. See Beneficials Appendix for solutions.

In our climate, certain insects are sure to become problematic if not handled preventively with the help of certain infrastructure:

Cucumber beetles and the damage they do by spreading bacterial wilt can be mostly avoided by installing insect screens in the openings of the roll ups. Highly recommended.

Spider mite populations are greatly reduced by keeping humidity levels in the greenhouse above 60%. We achieve this by adding fogging misters to the cucumber greenhouse. These are set on a simple watering clock that will have them turn on every hour (or half-hour on very hot days) for a few seconds.

Diseases are also very context-dependent. The first lines of defence is to use cultivars that are disease resistant. Using varieties that are resistant to both **downy and powdery mildew** is a must.

In our experience, fungal problems (**Alternaria** and **Cercospora** leaf spot) are more effectively treated by spraying biological-based fungicides, rather than other certifiable sulfate solutions. A bio-fungicide like Serenade (which works with *Bacillus subtilis*) contains living bacteria or fungi that help reduce pathogenic microbes by competition or antagonism. Such biological products don't harm beneficial insects and other non-target organisms. We often mix with copper for more effectiveness.

Pythium and the damage that it does to the transplants is often a result of soil that was not warm enough when the plants went into the ground. A product like Rootshield, a biological fungicide, can also be very effective in preventing pythium outbreak, especially with the first planting of the season.

Removing sick plants and cleaning your hands when handling them is also vital in protecting against bacterial or viral infections. Diseased plants should not be put in the compost, but in the garbage.

Another important practice is eliminating habitats suitable for insect survival. You do this by completely removing plant residues once the production is done. Planting a new succession directly after the other is not a good idea, hence the importance of having more than one greenhouse for continual production.

For anything specific, a good resource guide is your best bet. Vegetable MD online from Cornell University is one example of where to look.

POLLINATION

Most greenhouse cucumber varieties have been hybridized not to need pollination. The cultivars we use are parthenocarpic, and will develop fruits without the help of pollinators—this is why we can use screens to protect against cucumber beetles and other insects in the greenhouse.

WHEN IT GETS HOT

Cucumber plants will shut down and abort their fruits if temperatures stay above 86 °F (30 °C) for too long during the day. To help the plants cope on very hot days, these measures help:

- Ventilate 24h/7 and as much as possible.
- Roll-up sides and all openings should be open to their maximum
- Install shade cloth over the greenhouse

HARVEST

When production starts, cucumbers are harvested 3 times a week and in peak production, everyday. All the fruits that are about the size will be harvested and graded at the end of the harvest. To speed up the harvest we use tree planting bags. All damaged or split fruit is put aside during harvest.

We harvest with tree planting bags and clippers using both of our hands, clipping the fruits as close as possible to the main stem of the plants, then cutting the stem closer if necessary. English cukes should be 12-13" (30 cm) long and of uniform fullness throughout, and Lebanese cukes should be 5-6" (13 cm) long. If all is going well, you should be harvesting 2-3 fruits per plant/per week. Before storing the fruits, don't forget to mark the date of the harvest in order to practice effective rotation of stock.

STORAGE

Cucumbers are stored in a cold room kept at +/- 46F (8C)

APPENDIX 1: CALCULATING PLANT DENSITY

Just as we do when calculating spacing for greenhouse tomatoes, density for greenhouse cucumbers is measured by calculating the surface area that is planted and multiplying it by a density target, measured in square meters (1 meter = 3,3 feet). This is how you figure out in-row spacing. A standard density target for English cucumbers is 1,8 plants/m², while for Lebanese cucumbers, the density target should be 3 plants/m².

English cucumber example:

The greenhouse surface dedicated for cukes is 6M X 30M so 180 m²

Following our 42" (1.1M) bed set-up, cukes are planted every bed, for a total of 6 beds to be planted in that area.

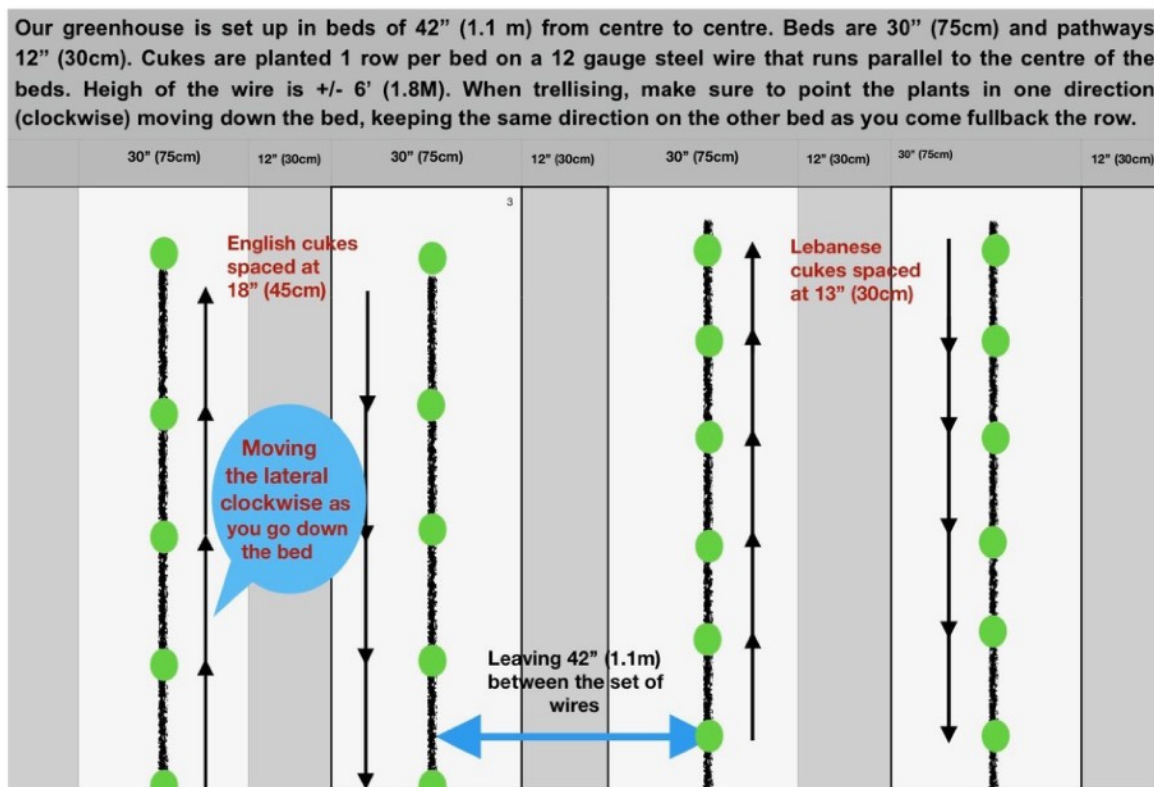
Aiming for a density of 1.8 heads/m², we multiply the surface area by the target; 180 m² X 1,8 heads/m² = 324 plants on 6 beds, thereby equating to 54 plants/bed.

When transplanting, there are 54 plants/30 metres, meaning 1 plant every 45 cm (18").

APPENDIX 2: CROP SPACING AND HARDWARE SET UP

Our greenhouse is set up in beds of 42" (1.1 m) from centre to centre. Beds are 30" (75 cm) and pathways 12" (30 cm). We plant cukes in tight single rows following a modified umbrella style of trellising.

The hardware for this set-up is 1 steel wires (minimum 12 gauges) set 42" (1.1M) apart above and in the middle of our 30" beds. The height of the wire should be



around 7' (2M) in order to be able to work on the plants without having to use ladders.

APPENDIX 3: FERTILITY

These are GENERAL target rates that will vary depending on your soil type, condition and growing areas. These can act as a starting point, but should be followed up with observations and soil tests.

The recipes are for 100' (30M) beds following the densities ranging from 1,8 to 3 plants/m². This is the weekly dose that is side-dressed and shallowly incorporated. Alternately, you can apply a double dose every two weeks.

Perform a soil salinity test on a weekly basis so that the fertilization can be adjusted to the growth speed.

Two weeks after transplanting:

Pelleted chicken manure or another 5-3-2 type fertilizer: 3.3 lbs (1.5 kg)

Compost : 5.5 lbs (2.5 kg)

Sul-Po-Mag: 0.66 lbs (0.3 kg)

Potassium sulfate: 1 lbs (0.5 kg)

Two weeks after harvest has begun:

Pelleted chicken manure or another 5-3-2 type fertilizer: 15 lbs (7 kg)

Compost: 7.5 lbs (3.5) kg

Sul-Po-Mag: 1 lbs (0.5 kg)

Potassium sulfate: 1.6 lbs (0.75 kg)

APPENDIX 4: DAILY CLIMATE CONTROL

Temperature control is the greenhouse grower's primary tool for adjusting the rate of development according to overall solar radiation and effects the balance between the

vegetative and generative growth. By playing with day and night temperature settings, you can steer the plant in a given direction.

Daytime temperature has an effect on cell elongation. A warm day temperature produces longer internodes and longer leaves. By decreasing this temperature, the opposite effect will be achieved. For the first part of the day, from sunrise until 9 or 10 o'clock, avoid excessively hot temperatures, which will cause wilting. Overall, the optimal temperature for photosynthesis is 72°F (22°C) $\pm$ 36°F (2°C).

Night temperature acts mainly on setting and respiration. The optimal temperature for setting is 64°F (18°C) $\pm$ 34°F (1°C). Night temperature is more often used to balance the desired $T^{\circ}24\text{h}$. Furthermore, the night T° is set according to the gap required with the day T° . A large gap between the two promotes generative growth, while little or no gap between daytime and nighttime temperatures encourages vegetative growth.

Average 24 hour temperature influences the speed at which new flower bunches and leaves are formed. It also acts on the speed of growth and the ripening of the fruit. $T^{\circ}24\text{h}$ affects the speed of fruit growth. Its increase over a sufficiently long period will accelerate fruit growth, but will lead to a decrease in fruit size.

APPENDIX 5: CUCUMBER PEST MANAGEMENT USING BENEFICIAL INSECTS

Pest and disease management in the greenhouse requires research, observation and careful timing. You need to be able to identify what pests you're dealing with, before they become an infestation. This is a useful basic guide for scouting your most common predators in the greenhouse.

Some important pest management strategies include:

- At the end of each season, remove all plant debris and wash down greenhouse interior as much as possible.
- Install screens on the sides of the greenhouse, under your rollups, to prevent cucumber beetles and other large insects from entering.
- Growing or buying banker plants to host parasitic wasps that feed on aphids - here is a very useful guide on how to grow your own banker plants
- Remove all weeds inside and around greenhouse perimeter - these can provide an easy habitat for thrips and other pests

- Hang yellow sticky cards throughout the crop (every 5 or 6 plants) to trap pests. These are used to monitor for whitefly, thrips and fungus gnats.
- Visual inspection of plant leaves. This should be done weekly for signs of two-spotted mites, aphid infestations and other problems - be sure to check the undersides of leaves, and inside flowers, where some pests congregate.

The following is a list of the most common greenhouse cucumber pests, and some biological controls used to combat them. Further information is available [here](#).

WHITE FLIES	White flies are small insects with broad wings covered with fine, snow-white waxy powder. Both adults and nymphs may feed on foliage by sucking juices from the underside of the leaf. They produce a honeydew which may result in a blackening of the leaf. Some species are also capable of transmitting certain viruses, which can greatly damage the plant.			
BENEFICIAL OR CONTROL	INTRODUCTION	QUANTITY	FREQUENCY	DURATION
Encarsia formosa	Preventive	1 per m ²	weekly or bi-weekly	5 introductions in total
Encarsia formosa is a tiny, 1 mm (1/20 inch) long, wasp that parasitizes immature stages of whitefly. As the wasp develops inside, greenhouse whitefly scales gradually turn black; parasitized sweet potato whitefly scales turn a tan colour.	Curative	5 per m ²	weekly	when over 80% of the whiteflies have been parasitized
	Hot spot	10 per m ²	weekly	when over 80% of the whiteflies have been parasitized

TWO-SPOTTED SPIDER MITES	Small, yellowish to dark green spider like pests that are the size of pepper flakes. They may be detected by dislodging them from the plant onto a piece of white paper and viewing them with a 10X magnifying lens. Webbing may be seen over the infested plants. Mites suck the sap from foliage. Leaves take on either a yellowish or bronze cast. During warm weather, spider mite populations reproduce very quickly and cause economic damage to cucumber plants. Controls should be applied at the first sign of spider mite infestation, even if the infestation is minimal.			
BENEFICIAL OR CONTROL	INTRODUCTION	QUANTITY	FREQUENCY	DURATION
Phytoseiulus persimilis	Curative light	5 per m2	weekly	weekly
Persimilis predatory mite that feeds on pest eggs, juveniles and adults. The predators develop twice as fast as the pest at moderate greenhouse temperatures, but it is almost entirely dependent on spider mites as food. Once their prey is depleted, predatory mite populations will decline in numbers.	Hot spot	100 per m2	weekly	continue releases until predator mites are present on all infested leaves.

FRUIT WORMS	Adult moths are yellowish-olive. Larvae vary in color from greenish-yellow, reddish brown or even black with paler stripes running lengthwise along the body. Fruit worms feed on the leaves and fruit and may bore into the stalk.			
BENEFICIAL OR CONTROL	INTRODUCTION	QUANTITY	FREQUENCY	DURATION
Bacillus thuringiensis Bt doesn't harm a majority of beneficial insects.	Curative light	at the first sign of eggs or infestation Apply the spray to the tops and bottoms of leaves, but only on infected plants.		as needed

APHIDS	Small, soft-bodied, pear-shaped insects that feed in colonies Infested leaves often curl and become distorted. Aphids can transmit virus diseases.			
BENEFICIAL OR CONTROL	INTRODUCTION	QUANTITY	FREQUENCY	DURATION
Ladybugs	Preventative	2 per m2	as needed	as needed
Common ladybugs will eat aphids at a rapid rate but will stay in the greenhouse only if aphids are present.	Curative medium	10 per m2	weekly or bi-weekly	as needed
	Hot spot	20 per m2	weekly or bi-weekly	as needed

THRIPS	Thrips damage leaves and distort the fruit of English cucumbers by feeding and laying eggs in leaf and flower tissues and young developing fruit. When thrip populations are high the adults are found in high numbers in the cucumber flowers. They can also spread viruses.			
BENEFICIAL OR CONTROL	INTRODUCTION	QUANTITY	FREQUENCY	DURATION
Cucumeris	Preventative	25/plant	weekly	ongoing
The predatory mite <i>Amblyseius cucumeris</i> feeds on immature stages of thrips. It is available in slow release bags, which are placed among plants as soon as plants begin to climb the trellis wire. <i>Amblyseius swirskii</i> is another option.	Curative	50/plant	weekly	until the percentage of leaves with predators is greater than that with thrips.